NALOXONE

Short-acting — re-narcotisation

Precipitates withdrawal

CHEMICAL NAME (5α)-4,5-epoxy-3,14-dihydroxy-17-(prop-2-en-1-yl)morphinan-6-one (hydrochloride)

CLASS Opioid receptor antagonist (antidote)

TRADE NAMES Narcan · Nyxoid (nasal) · Prenoxad (IM kit) · Naloxone HCl injection

THERAPEUTIC USES / INDICATIONS

- Opioid overdose / toxicity: respiratory depression, reduced consciousness.
- Reversal of post-operative opioid respiratory depression.
- Neonatal respiratory depression from maternal opioids (not if mother opioid-dependent).
- Diagnostic in unexplained coma; take-home naloxone for at-risk people.

ADMINISTRATION

- **IV** (preferred in hospital), **IM**, **SC**, **intranasal**, intraosseous.
- Titrate IV doses to restore adequate breathing (RR >10) — not full consciousness (avoid severe withdrawal) in dependent patients.
- Infusion for long-acting opioids (methadone, MR oxycodone).

DOSAGES

- **Opioid overdose (adult):** 400–2000 mcg IV, repeated every 2–3 min if needed; no response after 10 mg → reconsider diagnosis.
- **Post-op reversal (titrated):** 100–200 mcg IV, then 100 mcg every 2–3 min.
- **IM/SC:** if IV not possible (slower onset, longer action); **nasal:** 1.8 mg (Nyxoid, ≥14 yr), repeat after 2–3 min if needed.
- **Children:** overdose 10 mcg/kg IV, then 100 mcg/kg if no response; post-op 10–20 mcg/kg every 2–3 min.
- **Infusion:** 2/3 of initial effective dose per hour.

PHARMACOKINETICS

- **Onset:** IV 1–2 min; IM/SC 2–5 min; nasal 2–3 min. **Duration:** 30–90 min (shorter than most opioids).
- Metabolised in liver (glucuronidation). **t½:** 30–80 min.

NURSING MANAGEMENT

- Airway, breathing, oxygen, bag-valve-mask first; call for help.
- Monitor RR, SpO_2, GCS, pupils for at least 2–6 h after last dose (longer for long-acting opioids).
- Watch for withdrawal and agitation; ensure safety.
- Re-assess pain management plan.
- Provide take-home naloxone and training.

PHARMACODYNAMICS

- Competitive antagonist at **μ, κ, δ opioid receptors** → displaces opioids → reverses respiratory depression, sedation, miosis, hypotension.
- No effect in absence of opioids.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting.
- Tachycardia, hypertension.
- Sweating, tremor.

ADVERSE EFFECTS (SERIOUS)

- Acute opioid withdrawal in dependent patients: agitation, vomiting, diarrhoea, piloerection, aggression.
- Re-narcotisation when naloxone wears off (respiratory depression returns).
- Pulmonary oedema, arrhythmias, seizures (rare, post-op).
- Return of severe pain (post-operative/palliative).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** opioid dependence (titrate), cardiovascular disease, neonates of opioid-dependent mothers.

KEY INTERACTIONS

- Reverses opioid analgesia (and opioid effects of tramadol, codeine, loperamide in overdose).
- Buprenorphine reversal needs high/repeated doses.

PATIENT EDUCATION

- Effects wear off sooner than the opioid; call an ambulance even if the person wakes up.
- Carry naloxone if you or someone you know uses opioids; learn how to use it.
- Place person in recovery position after giving.

KEY POINTS

- Opioid antagonist — reverses respiratory depression.
- Short-acting; observe for re-narcotisation.
- 400 mcg IV, repeat/titrate; IM/nasal options.
- Precipitates withdrawal — titrate in dependence.

SOURCES CHECKED **[986]** NALOXONE 400 microgram/ml solution for injection or infusion (UK SmPC)
[987] Nyxoid 1.8 mg nasal spray, solution in a single-dose container (UK SmPC) **[988]** Naloxone Hydrochloride, IM, IV, SC (US label)

Full references p. 432–457